

Session-04: Bernoulli distribution

1. Assume that out of every 50 people in a city, one is a business owner. So, if one citizen is selected randomly, what is the distribution of business owners?

Solution: Given $p = \frac{1}{50}$ hence $q = \frac{49}{50}$

Let $X = 1$ represent that the person selected at random is a business owner.

and let $X = 0$ represent that the person selected is not a business owner.

Then, $P(X = 1) = p = \frac{1}{50} = 0.02$ and $P(X = 0) = 1 - p = 1 - \frac{1}{50} = \frac{49}{50} = 0.98$.

2. If one out of 15 stocks perform extraordinarily, then what is the performance of a stock randomly selected from the portfolio?

Solution: Let X be the Bernoulli random variable that denotes the performance of a stock.

Given $p = \frac{1}{15}$. Thus, $P(X = 1) = p = \frac{1}{15}$.

3. In a medical examination the chances of error are 15%. Find the Bernoulli distribution if one patient is randomly selected out of 60 patients.

Solution: Let X be the Bernoulli random that represents the number of correct reports.

It is given that 15% of 60 patients have error reports. This means that there are

$\frac{15\%}{60} = 9$ patients out of 60 who have received error reports. Thus, the number of patients receiving correct reports are $60 - 9 = 51$.

Thus, $P(X = 1) = \frac{51}{60} = 0.85$ and $P(X = 0) = \frac{9}{60} = 0.15$